

TCP — Congestion Control Algorithms

Network programming in python

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How to run the program:

We'll be using a Linux based operating system.

First, you'll need to download the following files and put them in the same directory:

1. Sender.py
2. Receiver.py

Then, open the integrated terminal and run the following commands:

1. Python3 Receiver.py

Open a second terminal and run the following commands:

1. Python3 Sender.py

Once the connection is completed and the sending of the file has started, you'll be asked if you'll want to resend the file 'Y' means yes, and 'N' means no.

To terminate the connection type 'N'.

Note: when checking for packet loss install the TC tool using the following command:

```
sudo apt install iproute
```

to see the packet loss, use the following command in the first terminal before the command for 'Receiver.py':

```
sudo tc qdisc change dev lo root netem loss XX%.
```

About the system:

In this project, we were instructed to write two program files Sender.py and Receiver.py which operate in the client server method (it's a model in networking computing system, which defines the relationship between cooperating programs. The server is a passive software that listens to the network and waits to receive requests. The client on the other hand, is activated by the user and turns to the server when he needs information or services from it).

In our case, the sender will send a message, a file sized at least 2MB. And the receiver will measure the time it took to get the said file.

The file will be sent in two halves, each part will be sent according to the CC algorithms learned in class.

The first half will be sent using the RENO algorithm, while the second half will be sent using the CUBIC algorithm.

In addition, we will use a packet loss tool called TC, and will monitor the results using Wireshark.

About the CC algorithms:

congestion on the internet happens when the computers send more packets to a particular link than that link can handle. If a link is receiving more packets than can fit on the wire, it will begin queueing up those packets, and if the link's queues fill up, the link will begin dropping those packets. Therefore we have the CC - Algorithm. In this program we use the reno and cubic CC Algorithms.

Cubic:

CUBIC is a standard TCP congestion control algorithm that uses a cubic function on the sender side to improve scalability and stability over fast and long-distance networks.

As we can see, the cubic function is very powerful- as x is lower the function grows very quickly, and then slows down as it approaches a particular point, and after crossing that specific point the function grows fairly quickly again.

The congestion window that has grown from the cubic function has the following properties:

1. Starts growing fast.
2. as the algorithm approaches the window at the last drop, the congestion window begins growing more slowly.
3. If the congestion window gets to the point at which drops occurred the last time, and does not experience a drop, begin growing slowly again, but then increase more quickly.

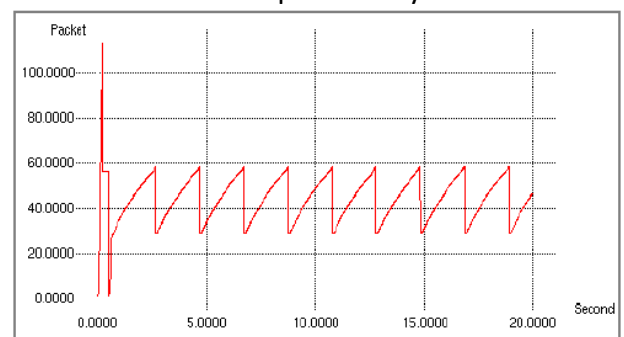
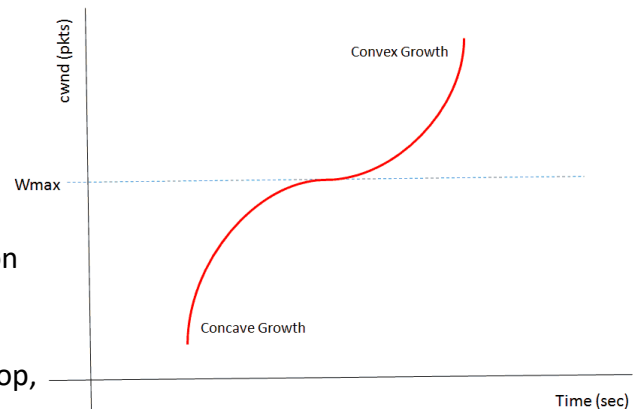
During the concave phase, the congestion window is catching up to where a packet was lost last time and slows down its growth to be fair to traditional TCP senders. If the algorithm gets there without experiencing drops, it can move into an "exploratory" phase, in which it grows quickly to discover newly available bandwidth.

Reno:

this algorithm operates by increasing the congestion window size exponentially until some threshold is reached.

After that threshold is reached, the function increases linearly. Once it experiences drops, it will aggressively drop the window size, and then enter "slow start" mode again, where it increases exponentially until it hits the threshold. Once the threshold is hit, it will increase linearly once again.

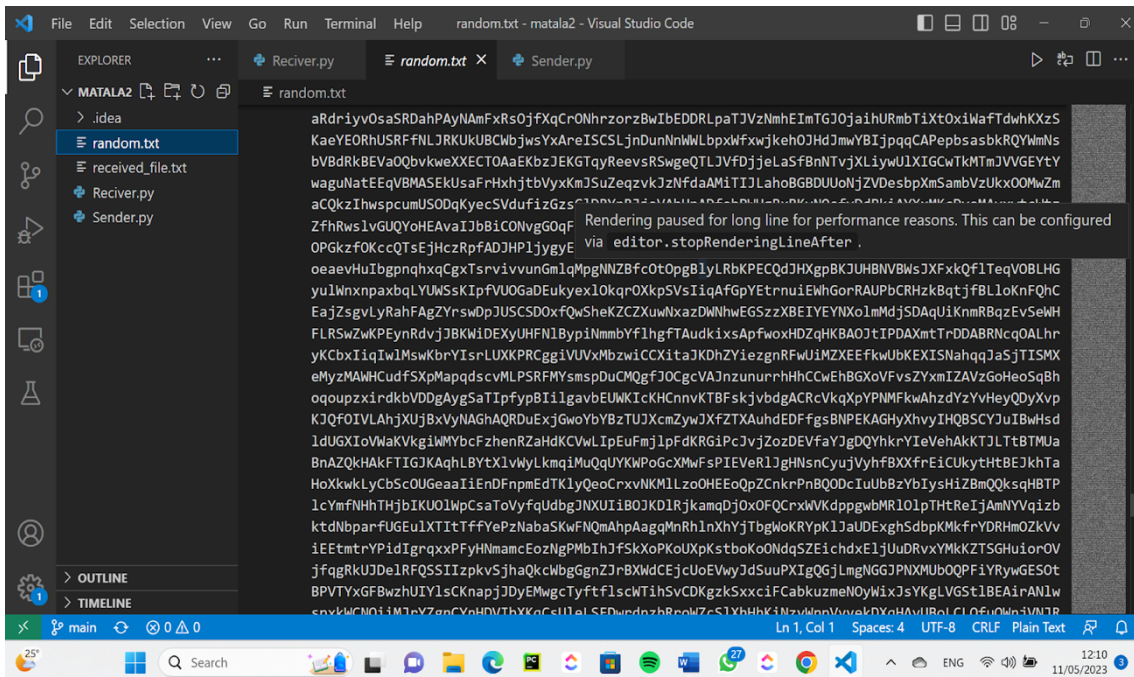
Since Reno grows linearly after crossing the slow start threshold, if a lot more bandwidths become available on a link, it'll take a long time for the algorithm to "discover" that available bandwidth.



Our code:

The main goal of our code is to create two objects of the classes sender and receiver. A terminal running the receiver file will run a short main program initialising a receiver object that acts as the server in the client server model. We then run a terminal that runs the sender file acting as the client in the server client model.

The server object is initialised and creates a random file of size between 3-6MB of chars. A file is created of random chars each line with 50 chars. As shown below:

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project named 'MATALA2' with files: '.idea', 'random.txt', 'received_file.txt', 'Receiver.py', and 'Sender.py'. The 'random.txt' file is selected and its content is displayed in the main editor. The text is a long, dense string of random characters, with a line break in the middle. A status bar at the bottom indicates 'Ln 1, Col 1', 'Spaces: 4', 'UTF-8', 'CRLF', and 'Plain Text'. The Windows taskbar is visible at the very bottom.

The receiver object creates a socket binds and listens.

Reciver.py:

we create a receiver object with a file name (for where to save what it receives), host , port and our id's for sending acks.

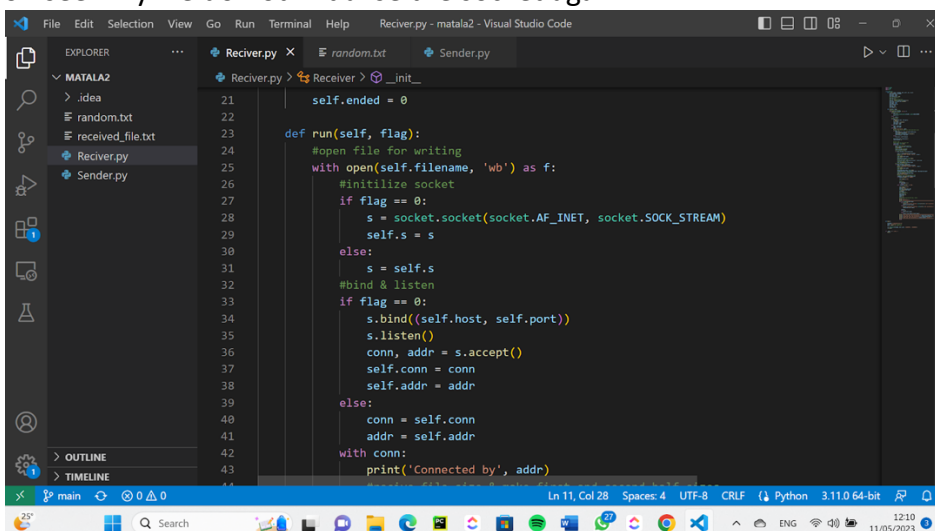
The instance of our receiver object then runs it function run() with a flag 0 indicating this is the first time its been made.

```
137 def main():
138
139     filename = "received_file.txt"
140     host = "127.0.0.1" # self ip
141     port = 9999 # port 9999 free
142
143     r1 = Receiver(filename, host, port, '332307073', '214329633')
144     r1.run(0)
145
146
147 if __name__ == "__main__":
148     main()
```

the constructor of our receiver class has extra fields such as a log for saving times, auth the XOR of our id's for sending acks and more variables that will help us later down the line.

```
class Receiver:
    def __init__(self, filename, host, port, id_1, id_2):
        self.filename = filename
        self.host = host
        self.port = port
        self.chunk_size = 1024
        #log for statistics
        self.log = []
        #ids for XOR auth operation
        self.xor = bin(int(id_1)^int(id_2))[2:]
        self.auth = str(self.xor).encode()
        self.conn = None
        self.addr = None
        self.filesize = 0
        self.s = None
        self.max = 0
        self.ended = 0
```

The run function opens a file to write into. If the flag is 0 meaning this is the first time we initialise a socket and save it to self.s and bind and listen we accept the connection and save the variables, if flag is not 0 we will later on see why we do not initialise the socket again.



With the connection we print connected to the address of the sender.

If this is our first time (flag = 0) the sender sends us the length of the file and we ack back and save it.

We then save the first_half size fo the file and keep track of total sent starting at 0.

Start a timer and enter the while loop.

```

41 class Reciver:
42     def __init__(self):
43         self.addr = self.addr
44         with conn:
45             print('Connected by', addr)
46             #receive file size & make first and second half sizes
47             if flag == 0:
48                 file_size_str = conn.recv(self.chunk_size)
49                 conn.sendall(self.auth)
50                 file_size = int(file_size_str.decode())
51                 self.filesize = file_size
52             else:
53                 file_size = self.filesize
54             first_half_size = file_size//2
55             total_recived = 0
56
57             #server loop
58             #start timer for first half sent
59             start_time = time.time()
60             while True:
61                 #receive data and write to file
62                 data = conn.recv(self.chunk_size)
63                 f.write(data)
64                 #total recived increased

```

In the while loop we receive chunks of the data and write it to the file, we update the total amount of data received. If the total amount received equals the first half size we send a ack and stop the timer and save the time to the log.

```

56 #server loop
57 #start timer for first half sent
58 start_time = time.time()
59 while True:
60     #receive data and write to file
61     data = conn.recv(self.chunk_size)
62     f.write(data)
63     #total recived increased
64     total_recived+=len(data)
65     #if finshed first half
66     if total_recived == first_half_size:
67         size = os.path.getsize(self.filename)
68         print(f'first half recived - size of : {size}')
69         #send auth
70         conn.sendall(self.auth)
71         #end timer
72         end_time_first_half = time.time()
73         #total first half time
74         first_half_time = end_time_first_half - start_time
75         #appending data to log
76         self.log.append(first_half_time)
77     #if second half finished
78     if total_recived == file_size:
79         #send ack

```

If the total amount received equals the size of the file, we send an ack and stop the timer. We append to the log the total time it took to receive the second half. And print file received. We then wait for the response of the sender if they want to resend or exit. If the decoded response is resend we recursively call the run function with flag 1. This will keep the connection open and work. Notice after we exit the recursive function we have another flag if it is 1 we break the while loop. This will allow us to exit gracefully, when closing the connection and closing the socket in the recursions.

```

76 #appending data to log
77 self.log.append(first_half_time)
78 #if second half finished
79 if total_recived == file_size:
80     size = os.path.getsize(self.filename)
81     print(f'second half recived - size of : {size}')
82     #send auth
83     conn.sendall(self.auth)
84     #end timer
85     end_second_half_time = time.time()
86     second_half_time = end_second_half_time - end_time_first_half
87     #append second half total time
88     self.log.append(second_half_time)
89     print('file recived')
90     #response for if to send again or end
91     response = conn.recv(self.chunk_size)
92     if response.decode() == 'RESEND':
93
94         conn.sendall(b'OK')
95
96     #rerun
97     self.run(1)
98     if self.ended == 1:
99         break

```

If the response decoded is bye, we ack and close the connection. We then wait $2 * \text{rtt}$ in our case we took the longest sending time of a half of a file to be careful. We then change the objects ended = 1 so that when we come back from the recursive functions we exit gracefully. (as opposed to being in a while loop trying to receive data from a closed connection and socket). We then compute the averages based on the values stored in self.log and print.

```

96 #rerun
97 self.run(1)
98 if self.ended == 1:
99     break
100 elif response.decode() == 'BYE':
101
102     conn.sendall(b'OK')
103     conn.close()
104     self.max = max(self.log)
105     time.sleep(2*self.max)
106     s.close()
107     self.ended = 1
108     #for statistics
109     avg_first = 0
110     avg_second = 0
111     count = 0
112     total = 0
113     #run through log and make avgs, totals
114     print(' ')
115     print('Statistics:')
116     print('*****')
117     for i in range(len(self.log)):
118         if i % 2 == 0:

```

The screenshot shows a Visual Studio Code editor window titled "Receiver.py - matala2 - Visual Studio Code". The Explorer sidebar on the left shows a project named "MATALA2" with files: ".idea", "random.txt", "received_file.txt", "Receiver.py", and "Sender.py". The main editor area displays the code for "Receiver.py", with line numbers 112 to 138 visible. The code is as follows:

```
112 print('*****')
113 print('*****')
114 for i in range(len(self.log)):
115     if i % 2 == 0:
116         print(f'first half iteration {int((i+1)/2)+1} sent succe
117         avg_first+=self.log[i]
118     else:
119         print(f'second half iteration {int(i/2)+1} sent successf
120         avg_second+=self.log[i]
121
122     count+=1
123     if i == 0:
124         total+=self.log[i]
125     else:
126         total+= self.log[i-1]+self.log[i]
127
128 print('*****')
129 print(f'first half avg: sent successfully in {avg_first/(count/2
130 print(f'second half avg: sent successfully in {avg_second/(count
131 print(f'total avg: sent successfully in {total/(count)} seconds.
132 print('*****')
133 break
134
135 def main():
136
137
138
```

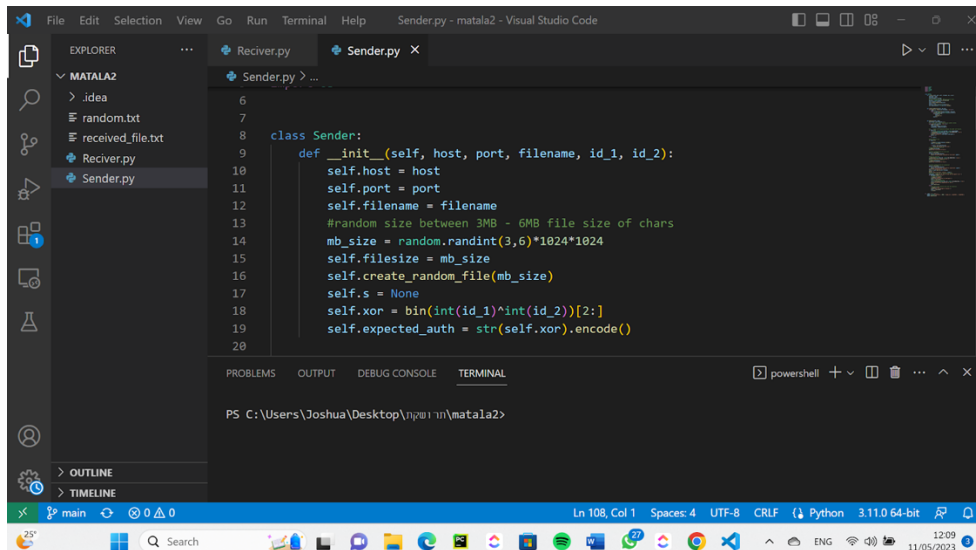
The status bar at the bottom indicates the current position is "Ln 11, Col 28", with "Spaces: 4", "UTF-8", "CRLF", and "Python 3.11.0 64-bit" settings.

Sender.py:

The main is run and a sender object is created with a host, port, filename and our id's, then its send_file function is run with a flag = 0.

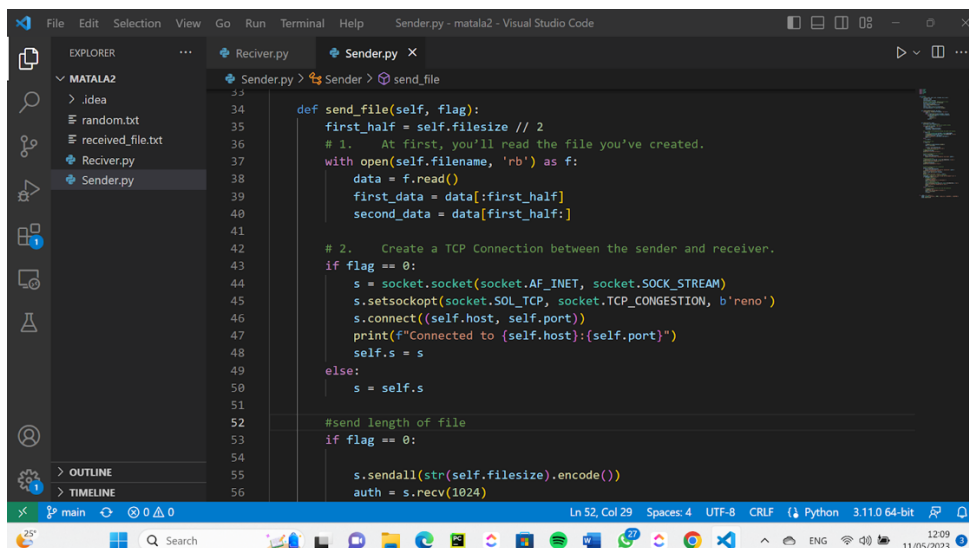
```
04
05 if __name__ == "__main__":
06     sender = Sender('127.0.0.1', 9999, 'random.txt', '332307073', '214329633')
07     sender.send_file(0)
08
```

The sender class constructor creates the object and creates a random size between 3-6MB then class create random file function.



The create random file function creates a random file the of size between 3-6MB and writes chars in lines of 50.

The sendfile function the is called with a flag 0. It opens the written file and reads the data and saves it in two halves to be sent. If the flag is 0 it creates a socket and connects to the receiver and saves the socket



If the flag is 0 send the length of the file to the receiver. Then send the first half and wait for ack from receiver. Print first half sent and change CC algo to cubic

```
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```

```
else:
    s = self.s

#send length of file
if flag == 0:

    s.sendall(str(self.filesize).encode())
    auth = s.recv(1024)

    if auth == self.expected_auth:
        print('Received ack on file size')
    # send first half of the file
    s.sendall(first_data)
    print("Sent first half of file:")

    # receive authentication for first half
    auth = s.recv(1024)
    if auth == self.expected_auth:
        print(f"Authentication received for first half: {auth}")

    #change CC algorithm
    s.setsockopt(socket.SOL_TCP, socket.TCP_CONGESTION, b'cubic')
    print("CC algorithm changed to cubic:")
    # send second half of the file
```

Ln 52, Col 29 Spaces: 4 UTF-8 CRLF Python 3.11.0 64-bit

Our findings running different levels of packet loss:

0% loss:

For 0 percent loss we sent the file once as shown

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc add dev lo root netem loss 0%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) █
```

The receiver received the file for the first time as shown:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 46184)
first half recieved - size of : 3141632
seoonnd half recieved - size of : 6287360
file recieved
█
```

We then sent the file again as shown:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc add dev lo root netem loss 0%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) █
```

The receiver received the file again for the second time as shown:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 receiver.py
Connected by ('127.0.0.1', 46184)
first half recived - size of : 3141632
seondnd half recived - size of : 6287360
file recieved
Connected by ('127.0.0.1', 46184)
first half recived - size of : 3141632
seondnd half recived - size of : 6287360
file recieved
█
```

We then sent the file again multiple times (more than or at least 5 times) and then exited

```
total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ █
```

The receiver received all the times the file was sent and printed this analysis: as one can see this is extremely fast and not surprising as there is 0% packet loss.

```
file recieved

Statistics:
*****
first half iteration 1 sent successfully in 0.007754325866699219 seconds.
second half iteration 1 sent successfully in 0.006215810775756836 seconds.
first half iteration 2 sent successfully in 0.03510260581970215 seconds.
second half iteration 2 sent successfully in 0.007299184799194336 seconds.
first half iteration 3 sent successfully in 0.013351202011108398 seconds.
second half iteration 3 sent successfully in 0.01095890998840332 seconds.
first half iteration 4 sent successfully in 0.010681390762329102 seconds.
second half iteration 4 sent successfully in 0.006400108337402344 seconds.
first half iteration 5 sent successfully in 0.012102603912353516 seconds.
second half iteration 5 sent successfully in 0.014298677444458008 seconds.
first half iteration 6 sent successfully in 0.01311182975769043 seconds.
second half iteration 6 sent successfully in 0.005118846893310547 seconds.
first half iteration 7 sent successfully in 0.009816646575927734 seconds.
second half iteration 7 sent successfully in 0.011287927627563477 seconds.
first half iteration 8 sent successfully in 0.009810209274291992 seconds.
second half iteration 8 sent successfully in 0.00483393669128418 seconds.
first half iteration 9 sent successfully in 0.010222196578979492 seconds.
second half iteration 9 sent successfully in 0.01852726936340332 seconds.
*****
first half avg: sent successfully in 0.01355033450656467 seconds.
second half avg: sent successfully in 0.009437852435641818 seconds.
total avg: sent successfully in 0.021958894199795194 seconds.
*****
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ █
```


We shall do this process for the following levels of loss: 10%, 15%, 20%, 25%, 30%

10% loss:

Sending file for the first time

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 10%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) █
```

Receiving file for the first time

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 52500)
first half recived - size of : 1568768
second half recived - size of : 3144986
file recived
█
```

Finished sending file multiple times to receiver

```
total file sent - size of : 3145728
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ █
```

Receiver printing stats: as we can see the process took slightly longer but still bearable.

```
file received

Statistics:
*****
first half iteration 1 sent successfully in 0.2158799171447754 seconds.
second half iteration 1 sent successfully in 1.121300220489502 seconds.
first half iteration 2 sent successfully in 0.08398771286010742 seconds.
second half iteration 2 sent successfully in 0.027344703674316406 seconds.
first half iteration 3 sent successfully in 0.03571462631225586 seconds.
second half iteration 3 sent successfully in 0.003339052200317383 seconds.
first half iteration 4 sent successfully in 0.21660518646240234 seconds.
second half iteration 4 sent successfully in 0.21417617797851562 seconds.
first half iteration 5 sent successfully in 0.048050642013549805 seconds.
second half iteration 5 sent successfully in 0.0029294490814208984 seconds.
first half iteration 6 sent successfully in 0.13849902153015137 seconds.
second half iteration 6 sent successfully in 0.21997570991516113 seconds.
first half iteration 7 sent successfully in 0.005781650543212891 seconds.
second half iteration 7 sent successfully in 0.20874285697937012 seconds.
first half iteration 8 sent successfully in 0.2299962043762207 seconds.
second half iteration 8 sent successfully in 0.005125761032104492 seconds.
first half iteration 9 sent successfully in 0.006598234176635742 seconds.
second half iteration 9 sent successfully in 0.004861593246459961 seconds.
*****
first half avg: sent successfully in 0.10901257726881239 seconds.
second half avg: sent successfully in 0.20086616939968532 seconds.
total avg: sent successfully in 0.3096086581548055 seconds.
*****
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

15% loss:

Sending file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 15%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Received ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n)
```

Receiving file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 60580)
first half recived - size of : 1568768
second half recived - size of : 3141632
file recived
```

Finished sending file multiple times to receiver:

```
total file sent - size of : 3145728
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 3145728
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

Receiver printing stats: here also the time has gotten slightly longer

We can see the first half sent via reno CC algo on average is doing better than second half using cubic CC algo

```
file recived

Statistics:
*****
first half iteration 1 sent successfully in 0.029301166534423828 seconds.
second half iteration 1 sent successfully in 0.019957780838012695 seconds.
first half iteration 2 sent successfully in 0.2243657112121582 seconds.
second half iteration 2 sent successfully in 0.48931217193603516 seconds.
first half iteration 3 sent successfully in 1.7009327411651611 seconds.
second half iteration 3 sent successfully in 1.8063304424285889 seconds.
first half iteration 4 sent successfully in 0.0961141586303711 seconds.
second half iteration 4 sent successfully in 0.0034341812133789062 seconds.
first half iteration 5 sent successfully in 0.11625838279724121 seconds.
second half iteration 5 sent successfully in 0.424685001373291 seconds.
first half iteration 6 sent successfully in 0.5196244716644287 seconds.
second half iteration 6 sent successfully in 0.3408691883087158 seconds.
first half iteration 7 sent successfully in 0.23624181747436523 seconds.
second half iteration 7 sent successfully in 0.6354207992553711 seconds.
first half iteration 8 sent successfully in 0.11029577255249023 seconds.
second half iteration 8 sent successfully in 0.009691953659057617 seconds.
first half iteration 9 sent successfully in 0.008368730545043945 seconds.
second half iteration 9 sent successfully in 0.9380323886871338 seconds.
*****
first half avg: sent successfully in 0.33794477250840926 seconds.
second half avg: sent successfully in 0.5186371008555094 seconds.
total avg: sent successfully in 0.8044689628813002 seconds.
*****
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

20% loss:

Sending file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 20%
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 4194304
Do you want to send the file again? (y/n) █
```

Receiving file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 33180)
first half recived - size of : 2093056
second half recived - size of : 4192714
file recived
█
```

Finished sending file to receiver multiple times:

```
total file sent - size of : 4194304
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 4194304
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 4194304
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ █
```

Receiver prints stats: we can see here that a big change has been made as now the files take seconds to send rather than milliseconds, notice we encounter outliers such as in the second last row of 19 seconds. We should consider taking this in to account when doing the averages.

```
first half recived - size of : 2096092
second half recived - size of : 4190878
file recived
```

Statistics:

```
*****
first half iteration 1 sent successfully in 4.7404820919036865 seconds.
second half iteration 1 sent successfully in 4.281578779220581 seconds.
first half iteration 2 sent successfully in 1.3726799488067627 seconds.
second half iteration 2 sent successfully in 0.4253201484680176 seconds.
first half iteration 3 sent successfully in 6.470535516738892 seconds.
second half iteration 3 sent successfully in 1.2220697402954102 seconds.
first half iteration 4 sent successfully in 0.2625274658203125 seconds.
second half iteration 4 sent successfully in 4.36931586265564 seconds.
first half iteration 5 sent successfully in 0.48596882820129395 seconds.
second half iteration 5 sent successfully in 0.752655029296875 seconds.
first half iteration 6 sent successfully in 4.721026420593262 seconds.
second half iteration 6 sent successfully in 1.1812093257904053 seconds.
first half iteration 7 sent successfully in 0.2794656753540039 seconds.
second half iteration 7 sent successfully in 0.9638276100158691 seconds.
first half iteration 8 sent successfully in 19.77590584754944 seconds.
second half iteration 8 sent successfully in 6.07660174369812 seconds.
*****
first half avg: sent successfully in 4.763573974370956 seconds.
second half avg: sent successfully in 2.4090722799301147 seconds.
total avg: sent successfully in 6.792858645319939 seconds.
*****
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

25% loss:

Sending file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 25%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n)
```

Receiving file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 49482)
first half recived - size of : 3145463
second half recived - size of : 6287941
file recived

```

Finished sending file multiple times to the receiver:

```
total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 6291456
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

Receiver prints stats: here we can see the average time is now above one minute, we see a awful outlier last line of 437 seconds and conclude that the difference between 10 - 15% is not the same as 20 - 25% we start to conclude exponential time in regards to level of loss %.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

file received
Connected by ('127.0.0.1', 49482)
first half received - size of : 3144367
seond half received - size of : 6291066
file received
Connected by ('127.0.0.1', 49482)
first half received - size of : 3143184
seond half received - size of : 6289989
file received

Statistics:
*****
first half iteration 1 sent successfully in 3.324810266494751 seconds.
second half iteration 1 sent successfully in 31.39530110359192 seconds.
first half iteration 2 sent successfully in 0.9808380603790283 seconds.
second half iteration 2 sent successfully in 18.945563793182373 seconds.
first half iteration 3 sent successfully in 99.49193859100342 seconds.
second half iteration 3 sent successfully in 4.474587917327881 seconds.
first half iteration 4 sent successfully in 2.3464291095733643 seconds.
second half iteration 4 sent successfully in 0.9952306747436523 seconds.
first half iteration 5 sent successfully in 4.9648237228393555 seconds.
second half iteration 5 sent successfully in 437.05942583084106 seconds.
*****
first half avg: sent successfully in 22.221767950057984 seconds.
second half avg: sent successfully in 98.57402186393738 seconds.
total avg: sent successfully in 77.08984723091126 seconds.
*****
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```


30% loss:

Sent file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ sudo tc qdisc change dev lo root netem loss 30%
[sudo] password for jodogo:
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 sender.py
Connected to 127.0.0.1:9999
Recived ack on file size
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 2097152
Do you want
```

Received file for the first time:

```
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ python3 reciver.py
Connected by ('127.0.0.1', 51164)
first half received - size of : 1044480
seondnd half received - size of : 2096781
file received
█
```

Finished sending file multiple times (5 times as this was taking forever):

```
Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 2097152
Do you want to send the file again? (y/n) y
Notified receiver to send again.
CC algorithm changed back to reno.

Sent first half of file:
Authentication received for first half: b'1000111000010110001010'
CC algorithm changed to cubic:
Sent second half of file:
Authentication received for second half: b'1000111000010110001010'
Sent second half of file:

total file sent - size of : 2097152
Do you want to send the file again? (y/n) n
Sent exit message to receiver.
jodogo@jod
ogo-Virtua
lBox:~/Des
ktop/Matal
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$ █
```

Receiver finished receiving the last time the file was sent and prints stats:

```
first half recived - size of : 1047834
second half recived - size of : 2095545
file recived
Connected by ('127.0.0.1', 51164)
first half recived - size of : 1047728
second half recived - size of : 2095386
file recived

Statistics:
*****
first half iteration 1 sent successfully in 1.3683812618255615 seconds.
second half iteration 1 sent successfully in 8.704511880874634 seconds.
first half iteration 2 sent successfully in 1.8507919311523438 seconds.
second half iteration 2 sent successfully in 13.570802450180054 seconds.
first half iteration 3 sent successfully in 5.474904298782349 seconds.
second half iteration 3 sent successfully in 2.3657453060150146 seconds.
first half iteration 4 sent successfully in 177.15200638771057 seconds.
second half iteration 4 sent successfully in 16.435860633850098 seconds.
first half iteration 5 sent successfully in 2.018491506576538 seconds.
second half iteration 5 sent successfully in 50.11139893531799 seconds.
first half iteration 6 sent successfully in 1330.7247593402863 seconds.
second half iteration 6 sent successfully in 630.7837238311768 seconds.
*****
first half avg: sent successfully in 253.09822245438895 seconds.
second half avg: sent successfully in 120.32867383956909 seconds.
total avg: sent successfully in 320.8615859746933 seconds.
*****
jodogo@jodogo-VirtualBox:~/Desktop/Matala2$
```

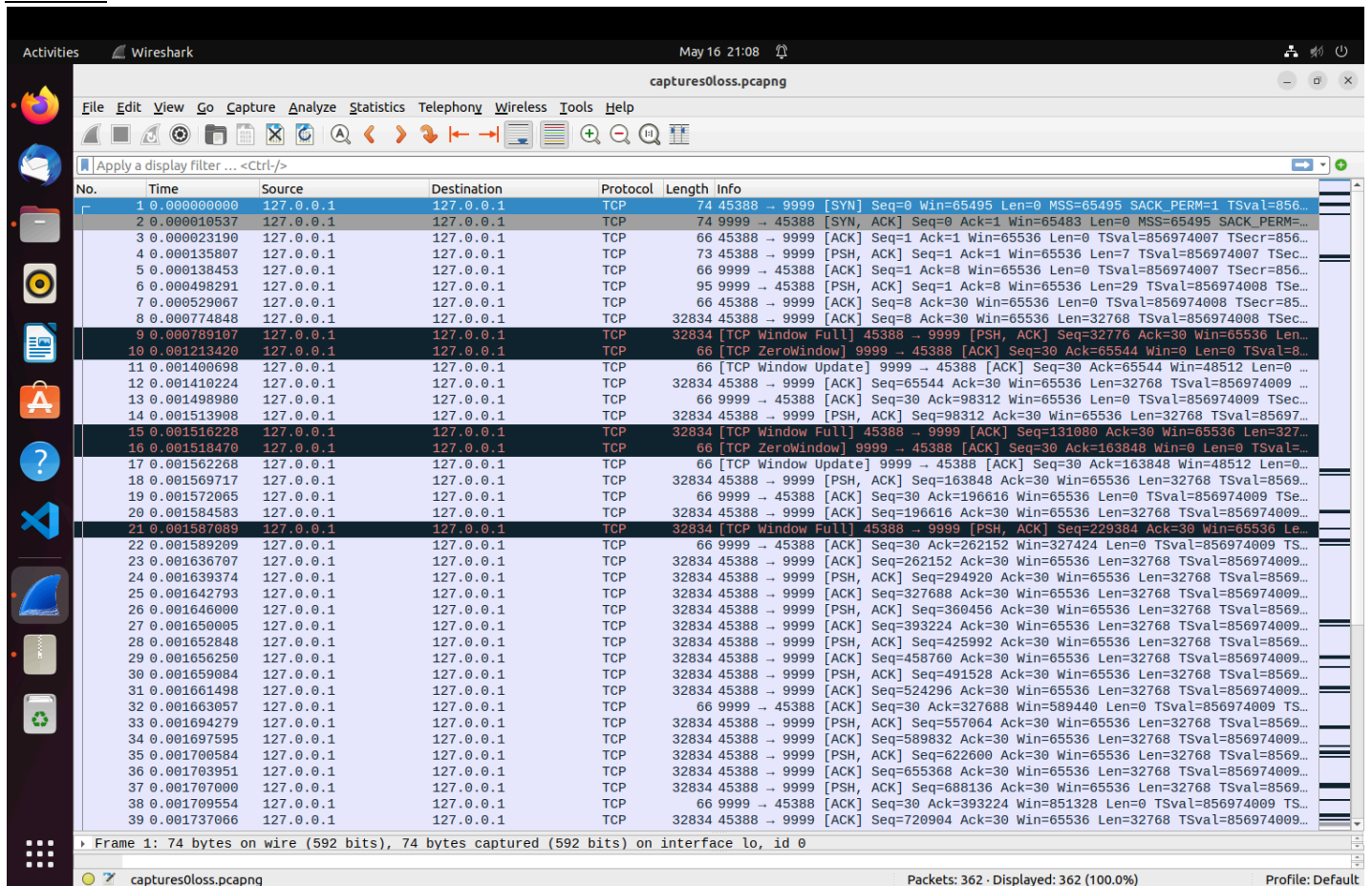

Key Findings:

- We set the receivers wait time to $2 \times$ (longest time a half was sent) to be careful as with high % level of packet loss the receiver would close to quickly. This then meant that the sender would close gracefully and the receiver would wait along time before printing the stats and then also exiting gracefully.
- We found that the time is exponential and the last rounds of loss level % 25 - 30 were much much longer as opposed to the 10-15% levels which were bearable.
- We used flags to ensure the TCP connection was only made once and stayed open until the sender decided to exit.
- We noticed outliers in the data that would effectively affect the averages in regards to times being printed by the receiver.
- We were receiving errors when closing the receiver connection and server and printed outputs to debug where things were going wrong. By using recursion we concluded that in a while loop listening and receiving data from the sender, that if we closed the connection and server of the receiver within a recursion call we had to make sure when it comes out of recursion that it exits gracefully, meaning that when using recursion to be careful that all resources are still available to use and if not to make a flag and exit gracefully.
- Our code would work Without loss of generality to the data being sent and decided to send a file of text.

Wireshark recordings:

in order to make it easier on the reader, in each packet loss screenshot, the connection between the sender and receiver will be at the top row in the first picture.

0% loss:



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	45388 → 9999 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM=1 TSval=856...
2	0.000010537	127.0.0.1	127.0.0.1	TCP	74	9999 → 45388 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM=...
3	0.000023190	127.0.0.1	127.0.0.1	TCP	66	45388 → 9999 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=856974007 TSecr=856...
4	0.000135807	127.0.0.1	127.0.0.1	TCP	73	45388 → 9999 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=7 TSval=856974007 TSecr=856...
5	0.000138453	127.0.0.1	127.0.0.1	TCP	66	9999 → 45388 [ACK] Seq=1 Ack=8 Win=65536 Len=0 TSval=856974007 TSecr=856...
6	0.000498291	127.0.0.1	127.0.0.1	TCP	95	9999 → 45388 [PSH, ACK] Seq=1 Ack=8 Win=65536 Len=29 TSval=856974008 TSecr=85...
7	0.000529067	127.0.0.1	127.0.0.1	TCP	66	45388 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=0 TSval=856974008 TSecr=85...
8	0.000774848	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=32768 TSval=856974008 TSecr=85...
9	0.000789107	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 45388 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=...
10	0.001213420	127.0.0.1	127.0.0.1	TCP	66	[TCP ZeroWindow] 9999 → 45388 [ACK] Seq=30 Ack=65544 Win=0 Len=0 TSval=8...
11	0.001400698	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 9999 → 45388 [ACK] Seq=30 Ack=65544 Win=48512 Len=0 ...
12	0.001410224	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=856974009 ...
13	0.001498980	127.0.0.1	127.0.0.1	TCP	66	9999 → 45388 [ACK] Seq=30 Ack=98312 Win=65536 Len=0 TSval=856974009 TSecr=85...
14	0.001513908	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=98312 Ack=30 Win=65536 Len=32768 TSval=85697...
15	0.001516228	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 45388 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=327...
16	0.001518470	127.0.0.1	127.0.0.1	TCP	66	[TCP ZeroWindow] 9999 → 45388 [ACK] Seq=30 Ack=163848 Win=0 Len=0 TSval=...
17	0.001562268	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 9999 → 45388 [ACK] Seq=30 Ack=163848 Win=48512 Len=0...
18	0.001569717	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=8569...
19	0.001572065	127.0.0.1	127.0.0.1	TCP	66	9999 → 45388 [ACK] Seq=30 Ack=196616 Win=65536 Len=0 TSval=856974009 TSecr=85...
20	0.001584583	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=196616 Ack=30 Win=65536 Len=32768 TSval=856974009...
21	0.001587089	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 45388 → 9999 [PSH, ACK] Seq=229384 Ack=30 Win=65536 Le...
22	0.001589209	127.0.0.1	127.0.0.1	TCP	66	9999 → 45388 [ACK] Seq=30 Ack=262152 Win=327424 Len=0 TSval=856974009 TS...
23	0.001636707	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=262152 Ack=30 Win=65536 Len=32768 TSval=856974009 TS...
24	0.001639374	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=294920 Ack=30 Win=65536 Len=32768 TSval=8569...
25	0.001642793	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=327688 Ack=30 Win=65536 Len=32768 TSval=856974009 TS...
26	0.001646000	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=360456 Ack=30 Win=65536 Len=32768 TSval=8569...
27	0.001650005	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=393224 Ack=30 Win=65536 Len=32768 TSval=856974009 TS...
28	0.001652848	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=425992 Ack=30 Win=65536 Len=32768 TSval=8569...
29	0.001656250	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=458760 Ack=30 Win=65536 Len=32768 TSval=856974009 TS...
30	0.001659084	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=491528 Ack=30 Win=65536 Len=32768 TSval=8569...
31	0.001661498	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=524296 Ack=30 Win=65536 Len=32768 TSval=856974009 TS...
32	0.001663957	127.0.0.1	127.0.0.1	TCP	66	9999 → 45388 [ACK] Seq=30 Ack=327688 Win=589440 Len=0 TSval=856974009 TS...
33	0.001694279	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=557064 Ack=30 Win=65536 Len=32768 TSval=8569...
34	0.001697595	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=589832 Ack=30 Win=65536 Len=32768 TSval=856974009 TS...
35	0.001700584	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=622600 Ack=30 Win=65536 Len=32768 TSval=8569...
36	0.001703951	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=655368 Ack=30 Win=65536 Len=32768 TSval=856974009 TS...
37	0.001707000	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [PSH, ACK] Seq=688136 Ack=30 Win=65536 Len=32768 TSval=8569...
38	0.001709554	127.0.0.1	127.0.0.1	TCP	66	9999 → 45388 [ACK] Seq=30 Ack=393224 Win=851328 Len=0 TSval=856974009 TS...
39	0.001737066	127.0.0.1	127.0.0.1	TCP	32834	45388 → 9999 [ACK] Seq=720904 Ack=30 Win=65536 Len=32768 TSval=856974009 TS...

The "[TCP Window Full]" message from Wireshark means that the system sending this TCP segment has filled up the receive window of the other end with the tcp segment in this packet.

Zero Window means that the receiver of the packets telling the sender to stop sending because there is no more buffer space for incoming packets. This is in almost all cases a sign of the receiver being too slow to process the incoming packets in time.

10% loss:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipp.tcp.local, "QM" question PTR _ipps.tcp.local, "QM" question
2	2.205807774	127.0.0.1	127.0.0.1	TCP	74	44722 → 9999 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=4246193155 TSecr=0 WS=128
3	2.205879274	127.0.0.1	127.0.0.1	TCP	74	9999 → 44722 [SYN, ACK] Seq=9 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=4246193159 TSecr=4246193155 WS=128
4	2.205888899	127.0.0.1	127.0.0.1	TCP	66	44722 → 9999 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=4246193155 TSecr=4246193155
5	2.206074857	127.0.0.1	127.0.0.1	TCP	73	44722 → 9999 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=7 TSval=4246193155 TSecr=4246193155
6	2.208514838	127.0.0.1	127.0.0.1	TCP	95	9999 → 44722 [PSH, ACK] Seq=1 Ack=8 Win=65536 Len=29 TSval=4246193158 TSecr=4246193155
7	2.208561785	127.0.0.1	127.0.0.1	TCP	66	44722 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=0 TSval=4246193158 TSecr=4246193158
8	2.208942654	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=32768 TSval=4246193158 TSecr=4246193158
9	2.208960358	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 44722 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246193158 TSecr=4246193158
10	2.209150883	127.0.0.1	127.0.0.1	TCP	66	[TCP ZeroWindow] 9999 → 44722 [ACK] Seq=30 Ack=65544 Win=0 Len=0 TSval=4246193158 TSecr=4246193158
11	2.209628505	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 9999 → 44722 [ACK] Seq=30 Ack=65544 Win=48512 Len=0 TSval=4246193159 TSecr=4246193158
12	2.209639362	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=4246193159 TSecr=4246193159
13	2.415594987	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 44722 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193159
14	2.415567370	127.0.0.1	127.0.0.1	TCP	78	9999 → 44722 [ACK] Seq=30 Ack=98312 Win=65536 Len=0 TSval=4246193365 TSecr=4246193365 SLE=65544 SRE=98312
15	2.415583730	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [PSH, ACK] Seq=98312 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
16	2.415672563	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 44722 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
17	2.415675689	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=30 Ack=131080 Win=48512 Len=0 TSval=4246193365 TSecr=4246193365
18	2.415796801	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=30 Ack=163848 Win=65536 Len=0 TSval=4246193365 TSecr=4246193365
19	2.415952951	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
20	2.415957000	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 44722 → 9999 [ACK] Seq=196016 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
21	2.415964256	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=30 Ack=196016 Win=196480 Len=0 TSval=4246193365 TSecr=4246193365
22	2.415989376	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [PSH, ACK] Seq=229384 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
23	2.415992257	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [ACK] Seq=262152 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
24	2.415994376	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [PSH, ACK] Seq=294920 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
25	2.415996420	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [ACK] Seq=327688 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
26	2.415998089	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=30 Ack=229384 Win=327424 Len=0 TSval=4246193365 TSecr=4246193365
27	2.416021221	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [PSH, ACK] Seq=360456 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
28	2.416023288	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [ACK] Seq=393224 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
29	2.416025415	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [PSH, ACK] Seq=425992 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
30	2.416027445	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [ACK] Seq=458760 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
31	2.416030624	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [PSH, ACK] Seq=491528 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
32	2.416032289	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=30 Ack=262152 Win=458496 Len=0 TSval=4246193365 TSecr=4246193365
33	2.416044371	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [ACK] Seq=524296 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
34	2.416050570	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 44722 → 9999 [ACK] Seq=5570315 Ack=150 Win=65536 Len=65483 TSval=4246197472 TSecr=4246193365
35	2.416041217	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=30 Ack=53920 Win=589440 Len=0 TSval=4246193365 TSecr=4246193365
36	2.416055984	127.0.0.1	127.0.0.1	TCP	32834	44722 → 9999 [PSH, ACK] Seq=622600 Ack=30 Win=65536 Len=32768 TSval=4246193365 TSecr=4246193365
37	2.416057423	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=30 Ack=327688 Win=720384 Len=0 TSval=4246193365 TSecr=4246193365

No.	Time	Source	Destination	Protocol	Length	Info
148	6.523311977	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=4849154 Ack=150 Win=65536 Len=65483 TSval=4246197472 TSecr=4246197472
149	6.523314416	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=150 Ack=4914637 Win=2814592 Len=0 TSval=4246197472 TSecr=4246197472
150	6.523332449	127.0.0.1	127.0.0.1	TCP	65549	[TCP Previous segment not captured] 44722 → 9999 [ACK] Seq=4980120 Ack=150 Win=65536 Len=65483 TSval=4246197472 TSecr=4246197472
151	7.160895318	127.0.0.1	127.0.0.1	TCP	65549	[TCP Retransmission] 44722 → 9999 [ACK] Seq=4914637 Ack=150 Win=65536 Len=65483 TSval=4246198118 TSecr=4246197472
152	7.160895112	127.0.0.1	127.0.0.1	TCP	78	[TCP ACKed unseen segment] 9999 → 44722 [ACK] Seq=150 Ack=5176569 Win=3112448 Len=0 TSval=4246198118 TSecr=4246197472 SLE=
153	7.160893508	127.0.0.1	127.0.0.1	TCP	914	[TCP Previous segment not captured] 44722 → 9999 [PSH, ACK] Seq=5242052 Ack=150 Win=65536 Len=848 TSval=4246198118 TSecr=4
154	7.160902773	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 152u1] [TCP ACKed unseen segment] 9999 → 44722 [ACK] Seq=150 Ack=5176569 Win=3112448 Len=0 TSval=4246198118 T
155	7.936017655	127.0.0.1	127.0.0.1	TCP	65549	[TCP Retransmission] 44722 → 9999 [ACK] Seq=5176569 Ack=150 Win=65536 Len=65483 TSval=4246198885 TSecr=4246198118
156	7.936058106	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=150 Ack=5242900 Win=3111296 Len=0 TSval=4246198885 TSecr=4246198885
157	7.937417996	127.0.0.1	127.0.0.1	TCP	95	9999 → 44722 [PSH, ACK] Seq=150 Ack=5242900 Win=3112448 Len=29 TSval=4246198887 TSecr=4246198885
158	7.938311455	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5242900 Ack=179 Win=65536 Len=65483 TSval=4246198887 TSecr=4246198887
159	7.938369848	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5368383 Ack=179 Win=65536 Len=65483 TSval=4246198887 TSecr=4246198887
160	7.939583500	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=5373866 Win=3079940 Len=0 TSval=4246198889 TSecr=4246198887
161	7.939653232	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5373866 Ack=179 Win=65536 Len=65483 TSval=4246198889 TSecr=4246198889
162	7.939676799	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5439349 Ack=179 Win=65536 Len=65483 TSval=4246198889 TSecr=4246198889
163	7.939774982	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=5504832 Win=3013504 Len=0 TSval=4246198889 TSecr=4246198889
164	7.940194508	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5504832 Ack=179 Win=65536 Len=65483 TSval=4246198889 TSecr=4246198889
165	7.983531737	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=5570315 Win=3112448 Len=0 TSval=4246198933 TSecr=4246198889
166	7.983594727	127.0.0.1	127.0.0.1	TCP	65549	[TCP Previous segment not captured] 44722 → 9999 [ACK] Seq=5635798 Ack=179 Win=65536 Len=65483 TSval=4246198933 TSecr=4246198933
167	7.983590673	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 165u1] 9999 → 44722 [ACK] Seq=179 Ack=5570315 Win=3112448 Len=0 TSval=4246198933 TSecr=4246198889 SLE=5635798
168	8.579160804	127.0.0.1	127.0.0.1	TCP	65549	[TCP Retransmission] 44722 → 9999 [ACK] Seq=5570315 Ack=179 Win=65536 Len=65483 TSval=4246199528 TSecr=4246198933
169	8.579189216	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=5701281 Win=3079940 Len=0 TSval=4246199528 TSecr=4246199528
170	8.579307994	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5701281 Ack=179 Win=65536 Len=65483 TSval=4246199528 TSecr=4246199528
171	8.579325494	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5766764 Ack=179 Win=65536 Len=65483 TSval=4246199528 TSecr=4246199528
172	8.579329792	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=5766764 Win=3046272 Len=0 TSval=4246199528 TSecr=4246199528
173	8.579614545	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=5832247 Win=3013504 Len=0 TSval=4246199528 TSecr=4246199528
174	8.579642445	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5832247 Ack=179 Win=65536 Len=65483 TSval=4246199529 TSecr=4246199528
175	8.579657968	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5897730 Ack=179 Win=65536 Len=65483 TSval=4246199529 TSecr=4246199528
176	8.579661764	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=5897730 Win=2800864 Len=0 TSval=4246199529 TSecr=4246199529
177	8.579679413	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=5963213 Win=2948096 Len=0 TSval=4246199529 TSecr=4246199529
178	8.579702828	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=5963213 Ack=179 Win=65536 Len=65483 TSval=4246199529 TSecr=4246199529
179	8.579713868	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=6028696 Ack=179 Win=65536 Len=65483 TSval=4246199529 TSecr=4246199529
180	8.579716737	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=6028696 Win=2914688 Len=0 TSval=4246199529 TSecr=4246199529
181	8.579738610	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=6094179 Ack=179 Win=65536 Len=65483 TSval=4246199529 TSecr=4246199529
182	8.579742159	127.0.0.1	127.0.0.1	TCP	66	9999 → 44722 [ACK] Seq=179 Ack=6159662 Win=2847872 Len=0 TSval=4246199529 TSecr=4246199529
183	8.579778470	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=6159662 Ack=179 Win=65536 Len=65483 TSval=4246199529 TSecr=4246199529
184	8.579789692	127.0.0.1	127.0.0.1	TCP	65549	44722 → 9999 [ACK] Seq=6225145 Ack=179 Win=65536 Len=65483 TSval=4246199529 TSecr=4246199529

Lost packets:

"tcp previous segment not captured" is a message created by Wireshark when it didn't see a packet that should have been in the trace.

Dup ACK means that you captured at the source of the data (not the receiving side). That is quite normal if the packet loss occurs somewhere in the path to the receiver.

TCP out-of-order means that particular frame was received in a different order from which it was sent (after a later packet in the sequence). It probably indicates that there are multiple paths between source and destination - and one travels a through a longer path.

TCP Retransmission occurs when the sender retransmits a packet after the expiration of the acknowledgement.

"TCP ACKed unseen segment" means that this packet acknowledges data that wasn't captured. It was transferred okay, and the receiver acknowledges it, but Wireshark can't find the packet in the capture. This usually happens when the capture device wasn't fast enough.

15% loss:

17	32.280577998	127.0.0.1	127.0.0.1	TCP	74	55282 → 9999 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=4246471907 TSecr=0 WS=128
18	32.280598189	127.0.0.1	127.0.0.1	TCP	74	9999 → 55282 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=4246471907 TSecr=4246471907 WS=128
19	32.280608561	127.0.0.1	127.0.0.1	TCP	66	55282 → 9999 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=4246471907 TSecr=4246471907
20	32.280659414	127.0.0.1	127.0.0.1	TCP	73	55282 → 9999 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=7 TSval=4246471907 TSecr=4246471907
21	32.280663624	127.0.0.1	127.0.0.1	TCP	66	9999 → 55282 [ACK] Seq=1 Ack=8 Win=65536 Len=0 TSval=4246471907 TSecr=4246471907
22	32.281394693	127.0.0.1	127.0.0.1	TCP	95	9999 → 55282 [PSH, ACK] Seq=1 Ack=8 Win=65536 Len=29 TSval=4246471909 TSecr=4246471907
23	32.281434330	127.0.0.1	127.0.0.1	TCP	66	55282 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=0 TSval=4246471908 TSecr=4246471908
24	32.281586588	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] [TCP Previous segment not captured] 55282 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
25	32.281590817	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 21#1] 9999 → 55282 [ACK] Seq=30 Ack=8 Win=65536 Len=0 TSval=4246471908 TSecr=4246471908 SLE=32776 SRE=65544
26	32.281718199	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 55282 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=32768 TSval=4246471908 TSecr=4246471908 SLE=32776 SRE=98312
27	32.281759638	127.0.0.1	127.0.0.1	TCP	66	[TCP ZeroWindowUpdate] 9999 → 55282 [ACK] Seq=30 Ack=65544 Win=0 Len=0 TSval=4246471908 TSecr=4246471908
28	32.282104769	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 9999 → 55282 [ACK] Seq=30 Ack=65544 Win=48512 Len=0 TSval=4246471909 TSecr=4246471909
29	32.282117485	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=4246471909 TSecr=4246471909
30	32.494211495	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 55282 → 9999 [PSH, ACK] Seq=30 Ack=65544 Win=48512 Len=0 TSval=4246472121 TSecr=4246472121 SLE=65544 SRE=98312
31	32.494222820	127.0.0.1	127.0.0.1	TCP	78	9999 → 55282 [ACK] Seq=30 Ack=98312 Win=65536 Len=0 TSval=4246472121 TSecr=4246472121 SLE=65544 SRE=98312
32	32.494242798	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=98312 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
33	32.494251088	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 55282 → 9999 [PSH, ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
34	32.494252493	127.0.0.1	127.0.0.1	TCP	66	9999 → 55282 [ACK] Seq=30 Ack=131080 Win=48512 Len=0 TSval=4246472121 TSecr=4246472121
35	32.494260335	127.0.0.1	127.0.0.1	TCP	66	9999 → 55282 [ACK] Seq=30 Ack=163848 Win=65536 Len=0 TSval=4246472121 TSecr=4246472121
36	32.494276027	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
37	32.494279455	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 55282 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121

No.	Time	Source	Destination	Protocol	Length	Info
40	32.494308985	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=262152 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
41	32.494311184	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=294920 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
42	32.494313182	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=327688 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
43	32.494316555	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 55282 → 9999 [PSH, ACK] Seq=327688 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
44	32.494317413	127.0.0.1	127.0.0.1	TCP	66	9999 → 55282 [ACK] Seq=30 Ack=229384 Win=327552 Len=0 TSval=4246472121 TSecr=4246472121
45	32.494336723	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=393224 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
46	32.494338894	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=425992 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
47	32.494341098	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 55282 → 9999 [PSH, ACK] Seq=491528 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
48	32.494342716	127.0.0.1	127.0.0.1	TCP	66	9999 → 55282 [ACK] Seq=30 Ack=262152 Win=458496 Len=0 TSval=4246472121 TSecr=4246472121
49	32.494359479	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 55282 → 9999 [PSH, ACK] Seq=557064 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
50	32.494361723	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=889832 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
51	32.494364011	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=522696 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
52	32.494367626	127.0.0.1	127.0.0.1	TCP	66	9999 → 55282 [ACK] Seq=30 Ack=327688 Win=720384 Len=0 TSval=4246472121 TSecr=4246472121
53	32.494399929	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=65368 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
54	32.494392881	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=688136 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
55	32.494394171	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=726904 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
56	32.494396458	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=753672 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
57	32.494397712	127.0.0.1	127.0.0.1	TCP	66	9999 → 55282 [ACK] Seq=30 Ack=360456 Win=757504 Len=0 TSval=4246472121 TSecr=4246472121
58	32.494405755	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=786448 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
59	32.494409073	127.0.0.1	127.0.0.1	TCP	78	9999 → 55282 [ACK] Seq=30 Ack=458760 Win=673824 Len=0 TSval=4246472121 TSecr=4246472121 SLE=491528 SRE=524296
60	32.494417033	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 55282 → 9999 [PSH, ACK] Seq=557064 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
61	32.494439279	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=884744 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
62	32.494441992	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=917512 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
63	32.494444410	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
64	32.494447065	127.0.0.1	127.0.0.1	TCP	32834	55282 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
65	32.494449198	127.0.0.1	127.0.0.1	TCP	86	[TCP Dup ACK 59#1] 9999 → 55282 [ACK] Seq=30 Ack=458760 Win=673824 Len=0 TSval=4246472121 TSecr=4246472121 SLE=557064 SRE=557064
66	32.494458899	127.0.0.1	127.0.0.1	TCP	32834	[TCP Out-Of-Order] 55282 → 9999 [ACK] Seq=458760 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
67	32.494460183	127.0.0.1	127.0.0.1	TCP	86	[TCP Dup ACK 59#2] 9999 → 55282 [ACK] Seq=30 Ack=458760 Win=673824 Len=0 TSval=4246472121 TSecr=4246472121 SLE=557064 SRE=557064
68	32.494671267	127.0.0.1	127.0.0.1	TCP	86	[TCP Dup ACK 59#3] 9999 → 55282 [ACK] Seq=30 Ack=458760 Win=673824 Len=0 TSval=4246472121 TSecr=4246472121 SLE=557064 SRE=557064
69	32.494683131	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 55282 → 9999 [ACK] Seq=524296 Ack=30 Win=65536 Len=32768 TSval=4246472121 TSecr=4246472121
70	32.494723079	127.0.0.1	127.0.0.1	TCP	86	9999 → 55282 [ACK] Seq=30 Ack=524296 Win=607488 Len=0 TSval=4246472121 TSecr=4246472121 SLE=951976 SRE=1015816 SLE=557064
71	32.499409056	127.0.0.1	127.0.0.1	TCP	95	[TCP ACKed unseen segment] 9999 → 55282 [PSH, ACK] Seq=30 Ack=1048584 Win=757504 Len=29 TSval=4246472126 TSecr=4246472126
72	32.499883251	127.0.0.1	127.0.0.1	TCP	65549	[TCP Previous segment not captured] 55282 → 9999 [ACK] Seq=114067 Ack=59 Win=65536 Len=65483 TSval=4246472126 TSecr=4246472126
73	32.499893833	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] [TCP ACKed unseen segment] 9999 → 55282 [ACK] Seq=59 Win=65536 Win=88848 Len=0 TSval=4246472126 TSecr=4246472126
74	32.499895807	127.0.0.1	127.0.0.1	TCP	65549	55282 → 9999 [ACK] Seq=1179556 Ack=59 Win=65536 Len=65483 TSval=4246472126 TSecr=4246472126
75	32.500084564	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 55282 [ACK] Seq=59 Win=65536 Win=1019392 Len=0 TSval=4246472127 TSecr=4246472127 SLE=111406
76	32.500086669	127.0.0.1	127.0.0.1	TCP	65549	55282 → 9999 [ACK] Seq=12295632 Ack=59 Win=65536 Len=65483 TSval=4246472127 TSecr=4246472126

20% loss:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000809909	127.0.0.1	127.0.0.1	TCP	74	43208 → 9999 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=4246646375 TSecr=0 WS=128
2	0.000927906	127.0.0.1	127.0.0.1	TCP	74	9999 → 43208 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=4246646375 TSecr=4246646375 WS=128
3	0.000943833	127.0.0.1	127.0.0.1	TCP	66	43208 → 9999 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=4246646375 TSecr=4246646375
4	0.0009318818	127.0.0.1	127.0.0.1	TCP	73	43208 → 9999 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=7 TSval=4246646375 TSecr=4246646375
5	0.0009323219	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=1 Ack=8 Win=65536 Len=0 TSval=4246646375 TSecr=4246646375
6	0.209833873	127.0.0.1	127.0.0.1	TCP	95	9999 → 43208 [PSH, ACK] Seq=1 Ack=8 Win=65536 Len=29 TSval=4246646375 TSecr=4246646375
7	0.210568972	127.0.0.1	127.0.0.1	TCP	66	43208 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=0 TSval=4246646585 TSecr=4246646585
8	0.210777609	127.0.0.1	127.0.0.1	TCP	32834	43208 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=32768 TSval=4246646586 TSecr=4246646586
9	0.212170695	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] 43208 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246646587 TSecr=4246646587
10	0.212323180	127.0.0.1	127.0.0.1	TCP	66	[TCP ZeroWindowUpdate] 9999 → 43208 [ACK] Seq=30 Ack=65544 Win=0 Len=0 TSval=4246646587 TSecr=4246646587
11	0.212502041	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 9999 → 43208 [ACK] Seq=30 Ack=65544 Win=48512 Len=0 TSval=4246646587 TSecr=4246646587
12	0.212539237	127.0.0.1	127.0.0.1	TCP	32834	43208 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=4246646587 TSecr=4246646587
13	0.212695743	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=30 Ack=98312 Win=65536 Len=0 TSval=4246646588 TSecr=4246646587
14	0.212809135	127.0.0.1	127.0.0.1	TCP	32834	43208 → 9999 [PSH, ACK] Seq=98312 Ack=30 Win=65536 Len=32768 TSval=4246646588 TSecr=4246646588
15	0.213159667	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=30 Ack=131080 Win=65536 Len=0 TSval=4246646588 TSecr=4246646588
16	0.213175383	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Full] [TCP Previous segment not captured] 43208 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=4246646588 TSecr=4246646588
17	0.213176026	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 16#1] 9999 → 43208 [ACK] Seq=30 Ack=131080 Win=65536 Len=0 TSval=4246646588 TSecr=4246646588 SLE=163848 SRE=163848
18	0.213191717	127.0.0.1	127.0.0.1	TCP	32834	[TCP Out-Of-Order] 43208 → 9999 [ACK] Seq=30 Ack=163848 Ack=30 Win=65536 Len=32768 TSval=4246646588 TSecr=4246646588
19	0.213198794	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=30 Ack=196616 Win=327424 Len=0 TSval=4246646588 TSecr=4246646588
20	0.213211231	127.0.0.1	127.0.0.1	TCP	32834	43208 → 9999 [ACK] Seq=196616 Ack=30 Win=65536 Len=32768 TSval=4246646588 TSecr=4246646588
21	0.230197596	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 43208 → 9999 [ACK] Seq=262152 Ack=30 Win=65536 Len=32768 TSval=4246646605 TSecr=4246646605
22	0.450309238	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 43208 → 9999 [ACK] Seq=196616 Ack=30 Win=65536 Len=32768 TSval=4246646825 TSecr=4246646588
23	0.450325709	127.0.0.1	127.0.0.1	TCP	86	9999 → 43208 [ACK] Seq=30 Ack=229384 Win=589440 Len=0 TSval=4246646825 TSecr=4246646825 SLE=196616 SRE=229384 SLE=262152
24	0.450361466	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 43208 → 9999 [PSH, ACK] Seq=229384 Ack=30 Win=65536 Len=32768 TSval=4246646825 TSecr=4246646825
25	0.450478793	127.0.0.1	127.0.0.1	TCP	32834	4320

No.	Time	Source	Destination	Protocol	Length	Info
49	0.450711185	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=30 Ack=726904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826
50	0.450733110	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 43208 → 9999 [PSH, ACK] Seq=753672 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
51	0.450740103	127.0.0.1	127.0.0.1	TCP	32834	43208 → 9999 [ACK] Seq=786440 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
52	0.450746174	127.0.0.1	127.0.0.1	TCP	32834	43208 → 9999 [PSH, ACK] Seq=819208 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
53	0.450751866	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 43208 → 9999 [PSH, ACK] Seq=884744 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
54	0.450754603	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 49#1] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=753672
55	0.450757400	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 49#2] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=753672
56	0.450778255	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 49#3] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=753672
57	0.450780992	127.0.0.1	127.0.0.1	TCP	86	[TCP Dup ACK 49#4] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=884744
58	0.450794574	127.0.0.1	127.0.0.1	TCP	32834	43208 → 9999 [ACK] Seq=917512 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
59	0.450801081	127.0.0.1	127.0.0.1	TCP	32834	[TCP Fast Retransmission] 43208 → 9999 [ACK] Seq=720904 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
60	0.450809477	127.0.0.1	127.0.0.1	TCP	32834	[TCP Out-Of-Order] 43208 → 9999 [ACK] Seq=851976 Ack=30 Win=65536 Len=32768 TSval=4246646826 TSecr=4246646826
61	0.450813091	127.0.0.1	127.0.0.1	TCP	86	[TCP Dup ACK 49#5] 9999 → 43208 [ACK] Seq=30 Ack=720904 Win=2238592 Len=0 TSval=4246646826 TSecr=4246646826 SLE=884744
62	0.450816948	127.0.0.1	127.0.0.1	TCP	78	9999 → 43208 [ACK] Seq=30 Ack=851976 Win=2139008 Len=0 TSval=4246646826 TSecr=4246646826 SLE=884744 SRE=950280
63	0.450821126	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=30 Ack=950280 Win=223904 Len=0 TSval=4246646826 TSecr=4246646826
64	0.450827644	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 43208 → 9999 [PSH, ACK] Seq=1015816 Ack=30 Win=65536 Len=32768 TSval=4246646845 TSecr=4246646845
65	0.450829603	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 43208 [ACK] Seq=30 Ack=950280 Win=2385568 Len=0 TSval=4246646845 TSecr=4246646826 SLE=1015816
66	0.450839834	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 43208 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4246646845 TSecr=4246646845
67	1.118549665	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 43208 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4246647493 TSecr=4246646845
68	1.118562118	127.0.0.1	127.0.0.1	TCP	86	9999 → 43208 [ACK] Seq=30 Ack=983048 Win=2517504 Len=0 TSval=4246647493 TSecr=4246647493 SLE=950280 SRE=983048 SLE=1015816
69	1.118577877	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 43208 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4246647493 TSecr=4246647493
70	1.329848704	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 43208 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4246647705 TSecr=4246647493
71	1.329858124	127.0.0.1	127.0.0.1	TCP	78	[TCP Previous segment not captured] 9999 → 43208 [ACK] Seq=30 Ack=1048584 Win=2648448 Len=0 TSval=4246647705 TSecr=4246647705 SLE=4246647705
72	1.333828743	127.0.0.1	127.0.0.1	TCP	95	[TCP Retransmission] 9999 → 43208 [PSH, ACK] Seq=30 Ack=1048584 Win=2648448 Len=29 TSval=4246647705 TSecr=4246647493
73	1.334413932	127.0.0.1	127.0.0.1	TCP	65549	43208 → 9999 [ACK] Seq=1048584 Ack=59 Win=65536 Len=65483 TSval=4246647709 TSecr=4246647709
74	1.334481196	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=59 Ack=1114067 Win=2779392 Len=0 TSval=4246647709 TSecr=4246647709
75	1.334570687	127.0.0.1	127.0.0.1	TCP	65549	43208 → 9999 [ACK] Seq=1114067 Ack=59 Win=65536 Len=65483 TSval=4246647709 TSecr=4246647709
76	1.334582591	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=59 Ack=1179550 Win=2910336 Len=0 TSval=4246647709 TSecr=4246647709
77	1.334620339	127.0.0.1	127.0.0.1	TCP	65549	43208 → 9999 [ACK] Seq=1179550 Ack=59 Win=65536 Len=65483 TSval=4246647709 TSecr=4246647709
78	1.334624282	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=59 Ack=1245033 Win=3041408 Len=0 TSval=4246647709 TSecr=4246647709
79	1.334914058	127.0.0.1	127.0.0.1	TCP	65549	43208 → 9999 [ACK] Seq=1245033 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647709
80	1.334976331	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=59 Ack=1310516 Win=3012804 Len=0 TSval=4246647710 TSecr=4246647710
81	1.334915649	127.0.0.1	127.0.0.1	TCP	65549	43208 → 9999 [ACK] Seq=1310516 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647709
82	1.334917642	127.0.0.1	127.0.0.1	TCP	66	9999 → 43208 [ACK] Seq=59 Ack=1375999 Win=2980224 Len=0 TSval=4246647710 TSecr=4246647710
83	1.334932846	127.0.0.1	127.0.0.1	TCP	65549	43208 → 9999 [ACK] Seq=1375999 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647709
84	1.334950613	127.0.0.1	127.0.0.1	TCP	65549	43208 → 9999 [ACK] Seq=1441482 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647710
85	1.334962413	127.0.0.1	127.0.0.1	TCP	65549	43208 → 9999 [ACK] Seq=1506965 Ack=59 Win=65536 Len=65483 TSval=4246647710 TSecr=4246647710

25% loss:

No.	Time	Source	Destination	Protocol	Length	Info
7	13.688583958	127.0.0.1	127.0.0.1	TCP	74	42566 → 9999 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=4246888581 TSecr=0 WS=128
8	13.688608063	127.0.0.1	127.0.0.1	TCP	74	9999 → 42566 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=4246888581 TSecr=4246888581 WS=128
9	13.688630365	127.0.0.1	127.0.0.1	TCP	66	42566 → 9999 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=4246888581 TSecr=4246888581
10	13.689015304	127.0.0.1	127.0.0.1	TCP	73	42566 → 9999 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=7 TSval=4246888582 TSecr=4246888581
11	13.689022458	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=1 Ack=8 Win=65536 Len=0 TSval=4246888582 TSecr=4246888582
12	13.898077013	127.0.0.1	127.0.0.1	TCP	95	9999 → 42566 [PSH, ACK] Seq=1 Ack=8 Win=65536 Len=29 TSval=4246888791 TSecr=4246888582
13	13.899474276	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=32768 TSval=4246888792 TSecr=4246888791
14	13.902693915	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=32776 Win=65536 Len=0 TSval=4246888795 TSecr=4246888792
15	14.126325087	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246888919 TSecr=4246888795
16	14.552479703	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246889445 TSecr=4246888795
17	15.402961229	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4246890235 TSecr=4246888795
18	15.402978395	127.0.0.1	127.0.0.1	TCP	78	9999 → 42566 [ACK] Seq=30 Ack=65544 Win=65536 Len=0 TSval=4246890236 TSecr=4246890235 SLE=32776 SRE=65544
19	15.402980820	127.0.0.1	127.0.0.1	TCP	32834	[TCP Window Update] 9999 → 42566 [ACK] Seq=30 Ack=294920 Win=65536 Len=0 TSval=4246891766 TSecr=4246891766
20	15.403008063	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 18#1] 9999 → 42566 [ACK] Seq=30 Ack=65544 Win=65536 Len=0 TSval=4246890236 TSecr=4246890236 SLE=93312 SRE=1
21	15.403020015	127.0.0.1	127.0.0.1	TCP	32834	[TCP Out-Of-Order] 42566 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=4246890236 TSecr=4246890236
22	15.403029914	127.0.0.1	127.0.0.1	TCP	66	[TCP Zero-Windows] 9999 → 42566 [ACK] Seq=30 Ack=131080 Win=0 Len=0 TSval=4246890236 TSecr=4246890236
23	16.024714933	127.0.0.1	127.0.0.1	TCP	66	[TCP Keep-Alive] 42566 → 9999 [ACK] Seq=131079 Ack=30 Win=65536 Len=0 TSval=4246890917 TSecr=4246890236
24	16.024846294	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 9999 → 42566 [ACK] Seq=30 Ack=131080 Win=65536 Len=0 TSval=4246890917 TSecr=4246890236
25	16.025042298	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4246890917 TSecr=4246890917
26	16.025070431	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=163848 Win=40512 Len=0 TSval=4246890918 TSecr=4246890917
27	16.057262020	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=4246891750 TSecr=4246890918
28	16.057279572	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=196616 Win=196608 Len=0 TSval=4246891750 TSecr=4246891750
29	16.057308219	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=196616 Ack=30 Win=65536 Len=32768 TSval=4246891750 TSecr=4246891750
30	16.057312852	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=229384 Ack=30 Win=65536 Len=32768 TSval=4246891750 TSecr=4246891750
31	16.057315762	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=262152 Win=458496 Len=0 TSval=4246891750 TSecr=4246891750
32	16.073247236	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 42566 → 9999 [PSH, ACK] Seq=306456 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
33	16.073261076	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 42566 [ACK] Seq=30 Ack=262152 Win=589448 Len=0 TSval=4246891766 TSecr=4246891766 SLE=306456 SRE=306456
34	16.073275756	127.0.0.1	127.0.0.1	TCP	32834	[TCP Out-Of-Order] 42566 → 9999 [ACK] Seq=262152 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
35	16.073281970	127.0.0.1	127.0.0.1	TCP	78	9999 → 42566 [ACK] Seq=30 Ack=294920 Win=65536 Len=0 TSval=4246891766 TSecr=4246891766 SLE=306456 SRE=393224
36	16.073295065	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=294920 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
37	16.073299027	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 42566 → 9999 [ACK] Seq=327688 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
38	16.073302842	127.0.0.1	127.0.0.1	TCP	78	9999 → 42566 [ACK] Seq=30 Ack=327688 Win=851456 Len=0 TSval=4246891766 TSecr=4246891766 SLE=306456 SRE=393224
39	16.073311176	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=393224 Win=982400 Len=0 TSval=4246891766 TSecr=4246891766
40	16.073322072	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=393224 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
41	16.073326537	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=425992 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
42	16.073327089	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=425992 Win=1096704 Len=0 TSval=4246891766 TSecr=4246891766
43	16.073335680	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=454760 Win=1096704 Len=0 TSval=4246891766 TSecr=4246891766

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No.	Time	Source	Destination	Protocol	Length	Info
46	16.873349377	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=491528 Win=1063936 Len=0 TSval=4246891766 TSecr=4246891766
47	16.873350756	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 46#1] 9999 → 42566 [ACK] Seq=30 Ack=491528 Win=1063936 Len=0 TSval=4246891766 TSecr=4246891766 SLE=524296 SRE=524296
48	16.873358767	127.0.0.1	127.0.0.1	TCP	32834	[TCP Out-Of-Order] 42566 → 9999 [PSH, ACK] Seq=491528 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
49	16.873360216	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=557064 Win=1030528 Len=0 TSval=4246891766 TSecr=4246891766
50	16.873369118	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 42566 → 9999 [ACK] Seq=589832 Ack=30 Win=65536 Len=32768 TSval=4246891766 TSecr=4246891766
51	16.873370214	127.0.0.1	127.0.0.1	TCP	78	[TCP Dup ACK 49#1] 9999 → 42566 [ACK] Seq=557064 Win=1030528 Len=0 TSval=4246891766 TSecr=4246891766 SLE=589832 SRE=589832
52	17.083430225	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=557064 Ack=30 Win=65536 Len=32768 TSval=4246891976 TSecr=4246891766
53	17.506954669	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 42566 → 9999 [PSH, ACK] Seq=622660 Win=1272648 Len=0 TSval=4246892393 TSecr=4246891976 SLE=557064 SRE=589832
54	17.506962381	127.0.0.1	127.0.0.1	TCP	78	9999 → 42566 [ACK] Seq=30 Ack=622660 Win=1272648 Len=0 TSval=4246892393 TSecr=4246891976 SLE=557064 SRE=589832
55	17.506962598	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=622660 Ack=30 Win=65536 Len=32768 TSval=4246892393 TSecr=4246892393
56	17.5069676205	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=655368 Ack=30 Win=65536 Len=32768 TSval=4246892393 TSecr=4246892393
57	17.5069679149	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=655368 Win=1358592 Len=0 TSval=4246892393 TSecr=4246892393
58	18.077759275	127.0.0.1	127.0.0.1	TCP	32834	[TCP Retransmission] 42566 → 9999 [ACK] Seq=655368 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892393
59	18.077775612	127.0.0.1	127.0.0.1	TCP	78	9999 → 42566 [ACK] Seq=30 Ack=688136 Win=1489536 Len=0 TSval=4246892970 TSecr=4246892970 SLE=65536 SRE=688136
60	18.077806329	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=688136 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
61	18.077825144	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=720984 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
62	18.077829339	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=720984 Win=1620480 Len=0 TSval=4246892970 TSecr=4246892970
63	18.077837936	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=753672 Win=1751552 Len=0 TSval=4246892970 TSecr=4246892970
64	18.077852478	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=753672 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
65	18.077860811	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=786448 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
66	18.077863134	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=786448 Win=1882496 Len=0 TSval=4246892970 TSecr=4246892970
67	18.077874226	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=819208 Win=2013440 Len=0 TSval=4246892970 TSecr=4246892970
68	18.077886410	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=819208 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
69	18.077892203	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=851976 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
70	18.077894557	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=851976 Win=2144384 Len=0 TSval=4246892970 TSecr=4246892970
71	18.077896151	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=884744 Win=2406272 Len=0 TSval=4246892970 TSecr=4246892970
72	18.077915831	127.0.0.1	127.0.0.1	TCP	32834	[TCP Previous segment not captured] 42566 → 9999 [ACK] Seq=917512 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
73	18.077920367	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4246892970 TSecr=4246892970
74	18.077922418	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 42566 [ACK] Seq=30 Ack=884744 Win=2406272 Len=0 TSval=4246892970 TSecr=4246892970 SLE=917512
75	18.077932789	127.0.0.1	127.0.0.1	TCP	32834	[TCP Out-Of-Order] 42566 → 9999 [PSH, ACK] Seq=884744 Ack=30 Win=65536 Len=32768 TSval=4246892971 TSecr=4246892970
76	18.077935894	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=983048 Win=2668288 Len=0 TSval=4246892971 TSecr=4246892971
77	18.077948539	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4246892971 TSecr=4246892971
78	18.077951719	127.0.0.1	127.0.0.1	TCP	66	9999 → 42566 [ACK] Seq=30 Ack=1015816 Win=2799232 Len=0 TSval=4246892971 TSecr=4246892971
79	18.077961589	127.0.0.1	127.0.0.1	TCP	32834	42566 → 9999 [PSH, ACK] Seq=1015816 Ack=30 Win=65536 Len=32768 TSval=4246892971 TSecr=4246892971
80	18.080149857	127.0.0.1	127.0.0.1	TCP	95	9999 → 42566 [PSH, ACK] Seq=30 Ack=1048584 Win=2930176 Len=29 TSval=4246892973 TSecr=4246892971
81	18.080606898	127.0.0.1	127.0.0.1	TCP	65549	[TCP Previous segment not captured] 42566 → 9999 [ACK] Seq=114067 Ack=30 Win=65536 Len=50483 TSval=4246892973 TSecr=4246892971
82	18.080678083	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 42566 [ACK] Seq=50 Ack=1048584 Win=3061120 Len=0 TSval=4246892973 TSecr=4246892971 SLE=114067

30% loss:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000000	127.0.0.1	127.0.0.1	TCP	74	69999 → 38838 [SYN, ACK] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=4247055751 TSecr=0 WS=128
2	0.000010402	127.0.0.1	127.0.0.1	TCP	74	69999 → 38838 [SYN, ACK] Seq=0 Ack=1 Win=65536 Len=0 MSS=65495 SACK_PERM TSval=4247055751 TSecr=4247055751 WS=128
3	0.000094369	127.0.0.1	127.0.0.1	TCP	73	38838 → 9999 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=7 TSval=4247055751 TSecr=4247055751
4	0.000106839	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=1 Ack=8 Win=65536 Len=0 TSval=4247055751 TSecr=4247055751
5	0.000228175	127.0.0.1	127.0.0.1	TCP	95	9999 → 38838 [PSH, ACK] Seq=1 Ack=8 Win=65536 Len=29 TSval=4247055751 TSecr=4247055751
6	0.000245021	127.0.0.1	127.0.0.1	TCP	66	38838 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=0 TSval=4247055751 TSecr=4247055751
7	0.000299967	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=8 Ack=30 Win=65536 Len=32768 TSval=4247055751 TSecr=4247055751
8	0.000462329	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=32776 Win=65536 Len=0 TSval=4247055751 TSecr=4247055751
9	0.000561587	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=32776 Ack=30 Win=65536 Len=32768 TSval=4247055751 TSecr=4247055751
10	0.000647554	127.0.0.1	127.0.0.1	TCP	78	9999 → 38838 [ACK] Seq=30 Ack=65544 Win=65536 Len=0 TSval=4247055751 TSecr=4247055751
11	0.000731942	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=65544 Ack=30 Win=65536 Len=32768 TSval=4247055752 TSecr=4247055751
12	0.000778258	127.0.0.1	127.0.0.1	TCP	32834 [TCP Window Full]	38838 → 9999 [PSH, ACK] Seq=88312 Ack=30 Win=65536 Len=32768 TSval=4247055752 TSecr=4247055751
13	0.000787289	127.0.0.1	127.0.0.1	TCP	66	[TCP ZeroWindow] 9999 → 38838 [ACK] Seq=30 Ack=131080 Win=0 Len=0 TSval=4247055752 TSecr=4247055752
14	0.000869455	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=131080 Win=48512 Len=0 TSval=4247055752 TSecr=4247055752
15	0.000916982	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4247055752 TSecr=4247055752
16	0.205805766	127.0.0.1	127.0.0.1	TCP	32834 [TCP Retransmission]	38838 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4247055957 TSecr=4247055752
17	0.414380155	127.0.0.1	127.0.0.1	TCP	32834 [TCP Retransmission]	38838 → 9999 [ACK] Seq=131080 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247055752
18	0.414409722	127.0.0.1	127.0.0.1	TCP	78	9999 → 38838 [ACK] Seq=30 Ack=163848 Win=65536 Len=0 TSval=4247056165 TSecr=4247056165 SLE=131080 SRE=163848
19	0.414461988	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=163848 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247056165
20	0.414491281	127.0.0.1	127.0.0.1	TCP	32834 [TCP Window Full]	38838 → 9999 [ACK] Seq=196616 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247056165
21	0.414496173	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=196616 Win=48512 Len=0 TSval=4247056165 TSecr=4247056165
22	0.414523035	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=229384 Win=65536 Len=0 TSval=4247056165 TSecr=4247056165
23	0.414743820	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=229384 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247056165
24	0.414754692	127.0.0.1	127.0.0.1	TCP	32834 [TCP Window Full]	38838 → 9999 [ACK] Seq=262152 Ack=30 Win=65536 Len=32768 TSval=4247056165 TSecr=4247056165
25	0.414773221	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=262152 Win=196488 Len=0 TSval=4247056166 TSecr=4247056165
26	0.414826137	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=294920 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
27	0.414834523	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=327688 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
28	0.414841739	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=360456 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
29	0.414848653	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=393224 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
30	0.414857660	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=294920 Win=327424 Len=0 TSval=4247056166 TSecr=4247056165
31	0.414908194	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=425992 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
32	0.414915599	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=458760 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
33	0.414922223	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=491528 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
34	0.414929485	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=524296 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
35	0.414941112	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=393224 Win=720384 Len=0 TSval=4247056166 TSecr=4247056166
36	0.415004361	127.0.0.1	127.0.0.1	TCP	32834 [TCP Previous segment not captured]	38838 → 9999 [ACK] Seq=589832 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
37	0.415011684	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=622608 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166

No.	Time	Source	Destination	Protocol	Length	Info
46	0.415112076	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=491528 Win=113216 Len=0 TSval=4247056166 TSecr=4247056166
47	0.415132617	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=950280 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
48	0.415139890	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=524296 Win=1244288 Len=0 TSval=4247056166 TSecr=4247056166
49	0.415147616	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1375232 Len=0 TSval=4247056166 TSecr=4247056166
50	0.415155740	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1506176 Len=0 TSval=4247056166 TSecr=4247056166 SLE=589832
51	0.415175339	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1637120 Len=0 TSval=4247056166 TSecr=4247056166 SLE=589832
52	0.415189371	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1768064 Len=0 TSval=4247056166 TSecr=4247056166 SLE=589832
53	0.415201596	127.0.0.1	127.0.0.1	TCP	86	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1899008 Len=0 TSval=4247056166 TSecr=4247056166 SLE=753672
54	0.415234392	127.0.0.1	127.0.0.1	TCP	32834 [TCP Retransmission]	38838 → 9999 [PSH, ACK] Seq=660180 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
55	0.415248943	127.0.0.1	127.0.0.1	TCP	32834 [TCP Retransmission]	38838 → 9999 [ACK] Seq=729904 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
56	0.415250151	127.0.0.1	127.0.0.1	TCP	94	[TCP Dup ACK 49#1] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1899008 Len=0 TSval=4247056166 TSecr=4247056166 SLE=884744
57	0.415272009	127.0.0.1	127.0.0.1	TCP	32834 [TCP Retransmission]	38838 → 9999 [ACK] Seq=851976 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
58	0.415279936	127.0.0.1	127.0.0.1	TCP	94	[TCP Dup ACK 49#2] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1899008 Len=0 TSval=4247056166 TSecr=4247056166 SLE=884744
59	0.415293772	127.0.0.1	127.0.0.1	TCP	94	[TCP Dup ACK 49#3] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1899008 Len=0 TSval=4247056166 TSecr=4247056166 SLE=884744
60	0.415629633	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [ACK] Seq=983048 Ack=30 Win=65536 Len=32768 TSval=4247056166 TSecr=4247056166
61	0.416095317	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=30 Ack=557064 Win=1906672 Len=0 TSval=4247056166 TSecr=4247056166 SLE=589832
62	0.626090947	127.0.0.1	127.0.0.1	TCP	32834 [TCP Retransmission]	38838 → 9999 [PSH, ACK] Seq=597064 Ack=30 Win=65536 Len=32768 TSval=4247056378 TSecr=4247056166
63	0.627826586	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=1015816 Win=2631616 Len=0 TSval=4247056378 TSecr=4247056378
64	0.835609417	127.0.0.1	127.0.0.1	TCP	32834	38838 → 9999 [PSH, ACK] Seq=1015816 Ack=30 Win=65536 Len=32768 TSval=4247056586 TSecr=4247056378
65	0.835772573	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=30 Ack=1048584 Win=2162560 Len=0 TSval=4247056587 TSecr=4247056586
66	1.041774360	127.0.0.1	127.0.0.1	TCP	95	9999 → 38838 [PSH, ACK] Seq=30 Ack=1048584 Win=2162560 Len=29 TSval=4247056793 TSecr=4247056586
67	1.041958005	127.0.0.1	127.0.0.1	TCP	65549 [TCP Previous segment not captured]	38838 → 9999 [ACK] Seq=1114067 Ack=59 Win=65536 Len=65483 TSval=4247056793 TSecr=4247056793
68	1.251146895	127.0.0.1	127.0.0.1	TCP	65549 [TCP Retransmission]	38838 → 9999 [ACK] Seq=1048584 Ack=59 Win=65536 Len=65483 TSval=4247056793 TSecr=4247056793
69	1.251188829	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=59 Ack=1179550 Win=2424448 Len=0 TSval=4247057002 TSecr=4247057002
70	1.251370060	127.0.0.1	127.0.0.1	TCP	65549 [TCP Previous segment not captured]	38838 → 9999 [ACK] Seq=1245623 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
71	1.251398819	127.0.0.1	127.0.0.1	TCP	65549 [TCP Previous segment not captured]	38838 → 9999 [ACK] Seq=1375999 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
72	1.251404413	127.0.0.1	127.0.0.1	TCP	78	[TCP Window Update] 9999 → 38838 [ACK] Seq=59 Ack=1179550 Win=2555392 Len=0 TSval=4247057002 TSecr=4247057002 SLE=124563
73	1.251442114	127.0.0.1	127.0.0.1	TCP	65549	38838 → 9999 [ACK] Seq=1441482 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
74	1.251446686	127.0.0.1	127.0.0.1	TCP	86	[TCP Window Update] 9999 → 38838 [ACK] Seq=59 Ack=1179550 Win=2817408 Len=0 TSval=4247057002 TSecr=4247057002 SLE=137595
75	1.251499574	127.0.0.1	127.0.0.1	TCP	65549 [TCP Out-Of-Order]	38838 → 9999 [ACK] Seq=1179550 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
76	1.251517698	127.0.0.1	127.0.0.1	TCP	65549 [TCP Out-Of-Order]	38838 → 9999 [ACK] Seq=1310516 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
77	1.251524577	127.0.0.1	127.0.0.1	TCP	78	9999 → 38838 [ACK] Seq=59 Ack=1310516 Win=2946176 Len=0 TSval=4247057002 TSecr=4247057002 SLE=137599 SRE=1506965
78	1.251550314	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=59 Ack=1506965 Win=2914688 Len=0 TSval=4247057002 TSecr=4247057002
79	1.251581019	127.0.0.1	127.0.0.1	TCP	65549	38838 → 9999 [ACK] Seq=1506965 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
80	1.251593022	127.0.0.1	127.0.0.1	TCP	65549	38838 → 9999 [ACK] Seq=1572448 Ack=59 Win=65536 Len=65483 TSval=4247057002 TSecr=4247057002
81	1.267476401	127.0.0.1	127.0.0.1	TCP	65549	38838 → 9999 [ACK] Seq=1637931 Ack=59 Win=65536 Len=65483 TSval=4247057018 TSecr=4247057002
82	1.310432542	127.0.0.1	127.0.0.1	TCP	66	9999 → 38838 [ACK] Seq=59 Ack=1704414 Win=3074720 Len=0 TSval=4247057061 TSecr=4247057018